PROJECT OVERVIEW

Power Glove Vision Project Overview

Architecture, controls, security, deployment, and project status.

2026 EDITION / IAIN BENNETT / MIT LICENSE

PowerGlove Vision lets you play RetroPie games by moving your hand in front of a camera connected to an Arduino UNO Q. Use your bare hand or a plain glove; there are no glove electronics to build. The UNO Q tracks your movements and sends controller input to a Raspberry Pi, where RetroArch sees a virtual gamepad named **PowerGlove Vision**.

The project includes eleven profiles: nine reusable Programs A-I and dedicated controls for Bad Street Brawler and Super Glove Ball. RetroPie can select a profile automatically when you launch a registered game. Glove Academy mode lets you practise without sending input to the cabinet. Its optional Pixel Pal-guided personalization wizard adjusts recognition to a player's hand without retraining the model, changing game mappings, or exposing raw thresholds during normal use.

The Dashboard's **Start controller** choice is retained across UNO Q application and system restarts as an armed preference. Armed does not mean that controls are always being sent: a registered RetroPie launch maintains a short renewable game session only while RetroArch is running. Controller delivery resumes after an UNO Q restart when that session is still live, then returns to neutral when the game ends, the session becomes stale, or an unregistered game is launched. **Stop controller** remains sticky until the player explicitly starts it again. Manual Dashboard profile selection remains available for testing outside the registered-game flow.

The cabinet supports two Super Glove Ball paths. `lr-fceumm` is the complete, standard-joystick fallback and remains the safe default. The separately named `lr-nestopia-powerglove` core supplies native absolute X/Y and Z coordinates plus open-hand, closed-hand/fist, and index-point states. Exact-ROM traces and deterministic headless tests confirm their packet bytes alongside detection, Start, four-direction activation/release, small continuous movement, and safe neutralization. Live cabinet sessions already confirm native Start, corrected Y orientation, controllable full-field X/Y movement, and low-lag stabilization; grab/throw, index fire, and fist-plus-forward Power Punch are now connected for cabinet validation. Native wrist rotation and the remaining action-button codes stay neutral until exact-ROM testing confirms them.

Choose a guide

User manuals

You want to...	Read...
Install both devices and play your first game	Installation Guide
Find a command or connection reminder	Quick Reference
Learn a game's gestures and try a short challenge	Game and gesture guide
Choose or experiment with Programs A-I	Programs A-I manual
Join Pixel Pal's suspiciously well-fingered scavenger hunt	Game and gesture guide and Programs A-I manual
Recognize matrix animations and letters	Matrix display guide

Technical documentation

You want to...	Read...
Get the complete project at a glance	Project overview PDF
Understand components and data flows	Architecture
Change settings or look up command flags	Configuration Reference
Review measured native and FCEUmm direction response	Direction-response benchmark
Understand network and pairing boundaries	Security policy
Change the project or its documentation	Contributing guide
Check dependency provenance or release history	Third-party components and Changelog

Pixel Pal's Extra-Digit Hunt

Pixel Pal has discovered that a few illustrated gloves left the art department with a generous interpretation of hand anatomy. Naturally, this is now a game.

Count every illustrated hand showing five fingers plus a thumb. Count each appearance, even when the same artwork returns. Pixel Pal has tucked the answers at the back of the two illustrated guides and behind a reveal in built-in Help, so there are no spoilers here.

An automated documentation check keeps the answers synchronized with the art. The art itself remains untouched in the interests of arcade archaeology - and because it is far too funny to fix.

Quick start

Prepare your UNO Q with Arduino App Lab and use an existing RetroPie installation. Connect both to the same trusted network and attach the camera through a powered USB hub. Keep a physical controller available for RetroArch setup.

The commands in the Installation Guide select the latest published stable release. Use the same release on both devices. Download `install-uno-q.sh` and `install-retropie.sh` from that published release. Run the first on the UNO Q and the second on RetroPie as your normal login user. Each verifies its package and requests sudo access when needed. The UNO installer includes the Arduino sketch, early-start helper, shutdown helper, and guarded USB-camera recovery helper; no separate App Lab import or helper installation is needed.

Follow the Installation Guide for copyable commands, pairing, calibration, and your first game. Both scripts also support `--check` and repeatable updates while preserving personal settings. The release-owned `config/profiles.json` baseline is backed up and replaced, while calibration and personal tuning under `data/` remain in place. Installer assets must be published before the release download commands become available.

The tested shared baseline includes responsive `0.28` activation and `0.14` release thresholds, full-camera-field native X/Y mapping with an 8% edge margin, and low-lag adaptive coordinate stabilization. These are suitable starting values for every installation. A neutral calibration is different: it records the palm center, apparent hand size, wrist angle, and resting jitter for one camera and playing position, so the installer never substitutes another person's recorded coordinates for yours.

Install the same release on the UNO Q and RetroPie. Automatic game-session resume depends on the current UNO Q worker and current RetroPie launch hook being present together; mixed old/new installations continue to fail safe but cannot provide the renewable session behavior.

When a registered Super Glove Ball ROM is present, the RetroPie installer offers to build the optional native core locally from pinned GPLv2 Nestopia source. If you decline, nothing changes and FCEUmm remains available. If you accept, both cores appear in RetroPie's per-ROM launch menu; FCEUmm stays selected until you choose the native entry. The project does not distribute ROMs or a compiled Nestopia core in its ordinary installation archive.

Controls

Calibration records the resting hand position that the app treats as the centre of movement. Move away from that position to give a direction and return to it to release that direction. Recalibrate after moving the camera or changing your playing position. Direction activation and release are shared by all FCEUmm profiles and automatically rise above measured resting-hand jitter; you do not normally calibrate each direction. Some profiles replace ordinary hand movement with wrist steering or other controls, as shown below.

Calibration accepts 24 clear hand observations at 70% confidence or better. Holding the same neutral pose at the same distance should reproduce a very similar reference, but ordinary tracking variation means the saved numbers will not be identical. A completed calibration is saved atomically and reused across games and restarts; an incomplete attempt does not replace the previous file.

Across the profiles, hold a **V sign** steadily for half a second to send Start and a **thumbs-up with the other fingers closed** to send Select. These poses suppress A/B attacks; some profiles can still generate directional or auxiliary input, so keep your hand near its resting position while using them. Start sends only one press and must see a clearly non-V pose for 0.30 seconds before it can trigger again.

Programs A-I

These reusable mappings produce ordinary NES controls. You do not need to launch Bad Street Brawler first. The table describes controller output; its effect depends on the game. A pulsed button repeatedly presses and releases.

Program	Movement	Actions and special gestures
A - Pinball	Ordinary movement is disabled.	Index curl sends A; thumb curl sends Up; wrist roll sends B. Pulling back toggles combined flippers.
B - Joust	Move your hand left or right.	Index or middle curl pulses A; thumb curl holds B.
C - Gyruss	Roll your wrist left or right.	A straight index finger holds A; pulling back sends B. Use the game's Attack Control B mode.
D - Challenge	All four hand-movement directions are reversed.	Thumb curl sends A; index curl sends B.
E - Defender II	Move your hand in four directions.	Thumb curl sends A; wrist roll sends B; ring-finger curl rapidly alternates left and right.

Program	Movement	Actions and special gestures
F - Sesame Street	Ordinary directional output is disabled.	Moving an open hand away from centre sends A; closing all fingers sends B.
G - Gun Smoke	Move your hand in four directions; wrist roll adds left or right.	Index curl sends A; pushing forward sends B. Combine them for A+B. Menu guard suppresses all ordinary controller output.
H - General	Move your hand in four directions.	Index curl pulses A; thumb curl pulses B.
I - Knight Rider	Roll your wrist to steer; lower your hand to brake.	Index curl sends Up for acceleration; pushing sends Up+A for turbo; thumb curl sends B.

Programs A, D, and H have no default ROM assignment. Choose one on Dashboard to try it, then register the exact game filename if you want automatic selection. The Programs manual includes illustrations; the Game and gesture guide adds objectives and tips.

Bad Street Brawler

Gesture	Controller output
Move your hand left, right, up, or down	Corresponding D-pad direction
Curl your thumb	Pulsed B
Curl your middle finger	A+B
Roll your wrist left or right	A plus that direction
Push toward the camera	Glove Zap: short simultaneous Left + Right pulse

Push toward the camera for Glove Zap, then return to your starting distance before trying again. A push or pull must cross its threshold on two consecutive fresh observations and travel at least 0.10 palm-scale units in the intended direction within 250 ms. This rejects a one-frame scale jump and a stationary hand that merely begins near or far from the camera. Bad Street Brawler needs its game-specific emulator setting; see the configuration reference.

Super Glove Ball

Gesture	Controller output
Move your hand left, right, up, or down	Corresponding D-pad direction
Curl your index finger	A
Curl your thumb	B

The same responsive mapping remains the explicit FCEUmm fallback. Headless tests using the same exact ROM show that both paths visibly activate and release every direction by frame 3. Their semantics differ: FCEUmm supplies held digital directions, while the native core supplies an absolute target position. The native path has passed exact-ROM detection, Start, continuous X/Y, absolute Z, open/fist/index packet, and safe-neutralization tests. The three native hand states are connected for grab/throw, index fire, and fist-plus-forward Power Punch cabinet validation. Wrist rotation and remaining native action-button codes stay neutral. See the native compatibility record.

The eight-ROM input audit confirms that the other listed games consume standard NES controller bits. They continue to use FCEUmm and the same global recognition settings.

Use the web interface

Page	What it does
Dashboard, `/dashboard`	Shows the camera and generated inputs; selects the current profile and starts or stops delivery.
Glove Academy, `/learn`	Provides sixteen mapping-independent practice lessons and guided gesture tuning, with game input paused.
Help, `/help`	Opens the local manuals and PDFs; **This cabinet** shows current connection details.
Setup, `/setup`	Saves connection, camera, and startup settings; the Games section edits RetroPie mappings with backup and restore. Pairing requires HTTPS on port 8443.

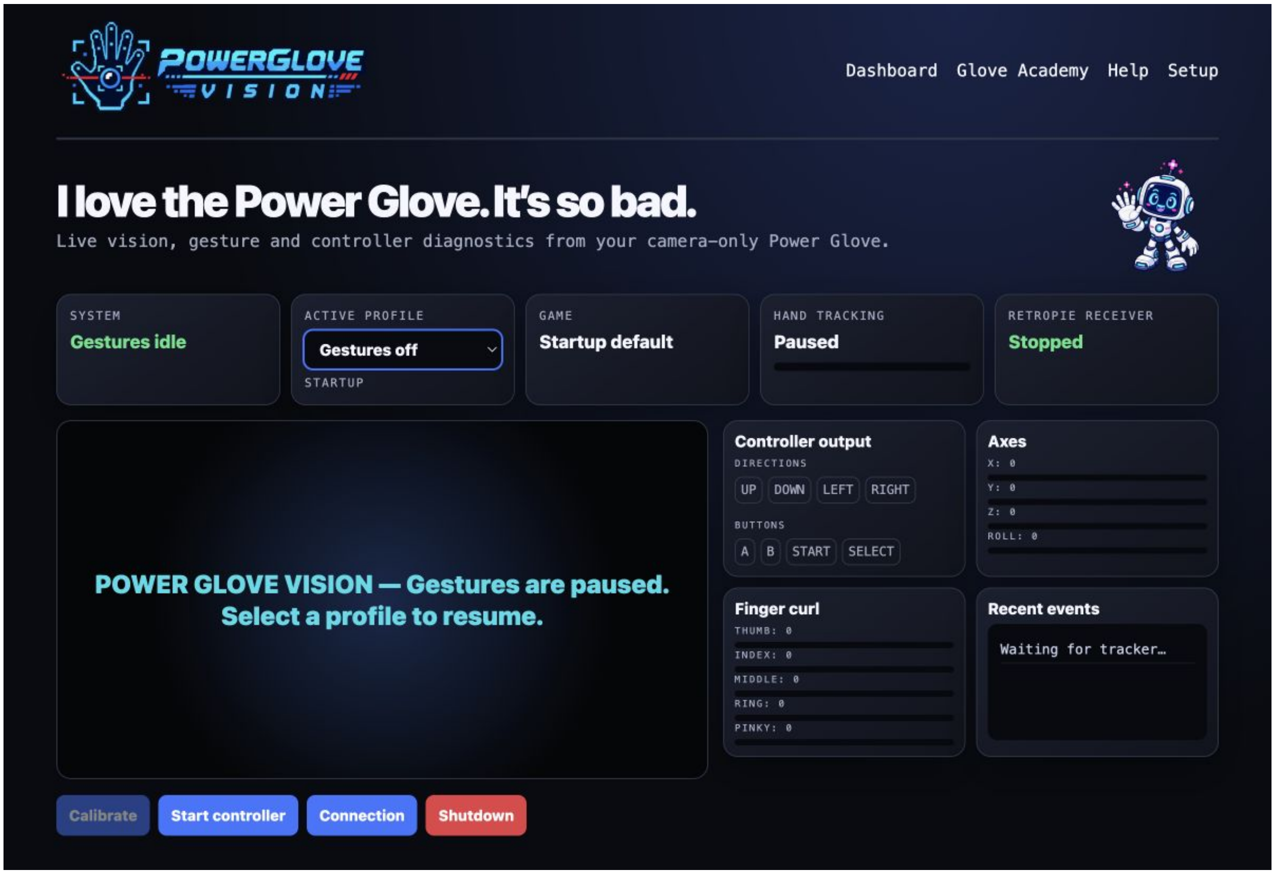
With **Gestures off** selected, the camera stays closed. Choose an active profile or open Glove Academy to begin. Wait for the camera view before practicing or playing; starting immediately after a reboot can take longer.

The live camera is diagnostic rather than part of controller output. Camera capture continuously keeps only the newest frame, and browser JPEG encoding runs on a separate latest-preview worker that may drop stale preview jobs. The preview remains capped at 5 fps. Closing Dashboard or Glove Academy still avoids optional drawing and encoding work; tracking and controller delivery continue.

Strong light behind the player can leave the hand dark even when the room looks bright. Prefer light from the camera side or move bright windows out of the background. The project does not force hardware backlight compensation: on the tested Razer Kiyo Pro it made the measured backlit scene darker, and aggressive manual exposure can trade brightness for motion blur and reduced frame rate.

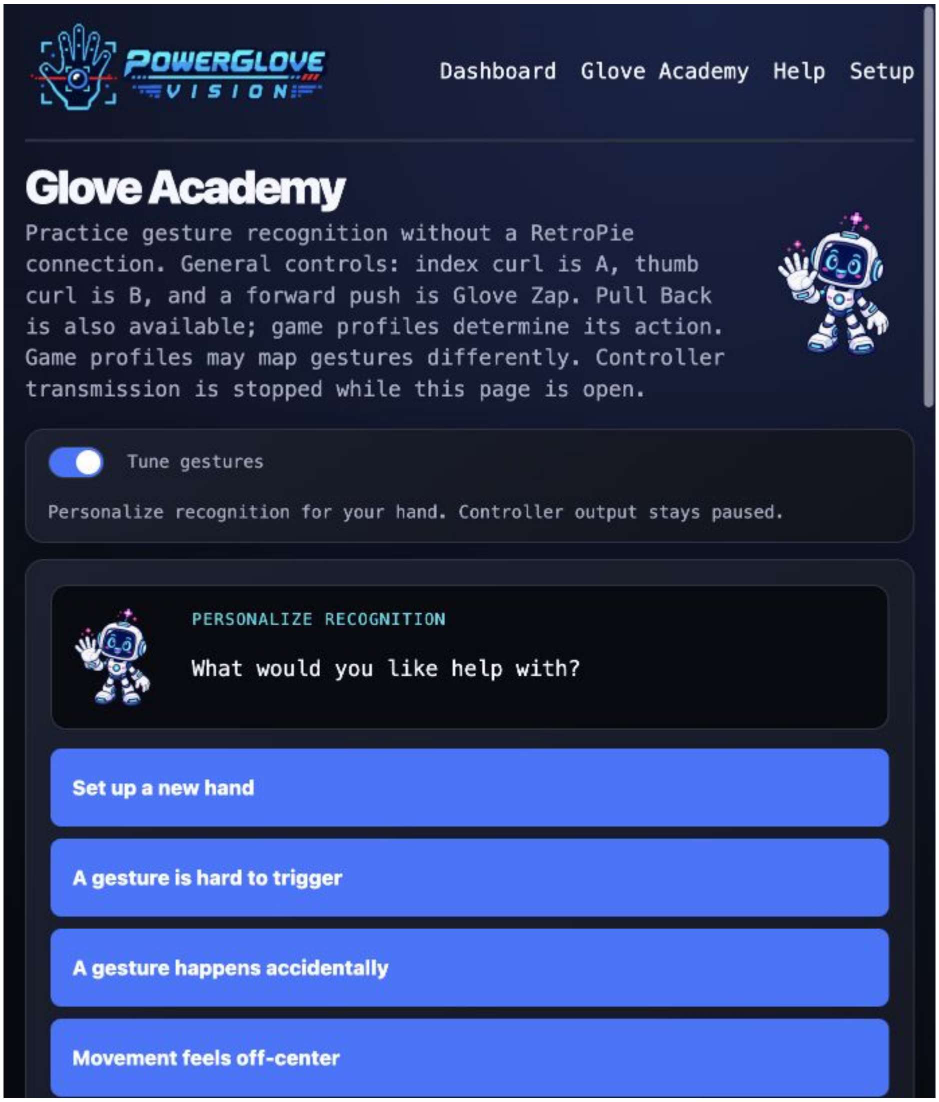
The UNO Q host helper supports one UVC camera. Installation works with or without the camera connected. On the first healthy sighting it records the camera and its actual parent USB hub in a root-owned allowlist, disables autosuspend for both, and automatically updates that association if the camera is later moved to a different hub. If the camera remains missing for 15 seconds while vision is requested, PowerGlove Vision makes one guarded recovery attempt for that outage by resetting only the last successfully observed hub. This can briefly interrupt USB Ethernet; Wi-Fi remains available. If a camera has never been seen-or does not return after that attempt-reconnect or power-cycle it and check the hub and cable rather than repeatedly resetting it.

Stop controller pauses delivery while leaving active tracking available. **Gestures off** closes the camera. **Shutdown** requests a Linux halt, but the tested UNO Q restarts afterward. A disappearing website is not proof that it is safe to remove power. See the installation guide before using Shutdown.



Dashboard showing the selected profile and controller readings

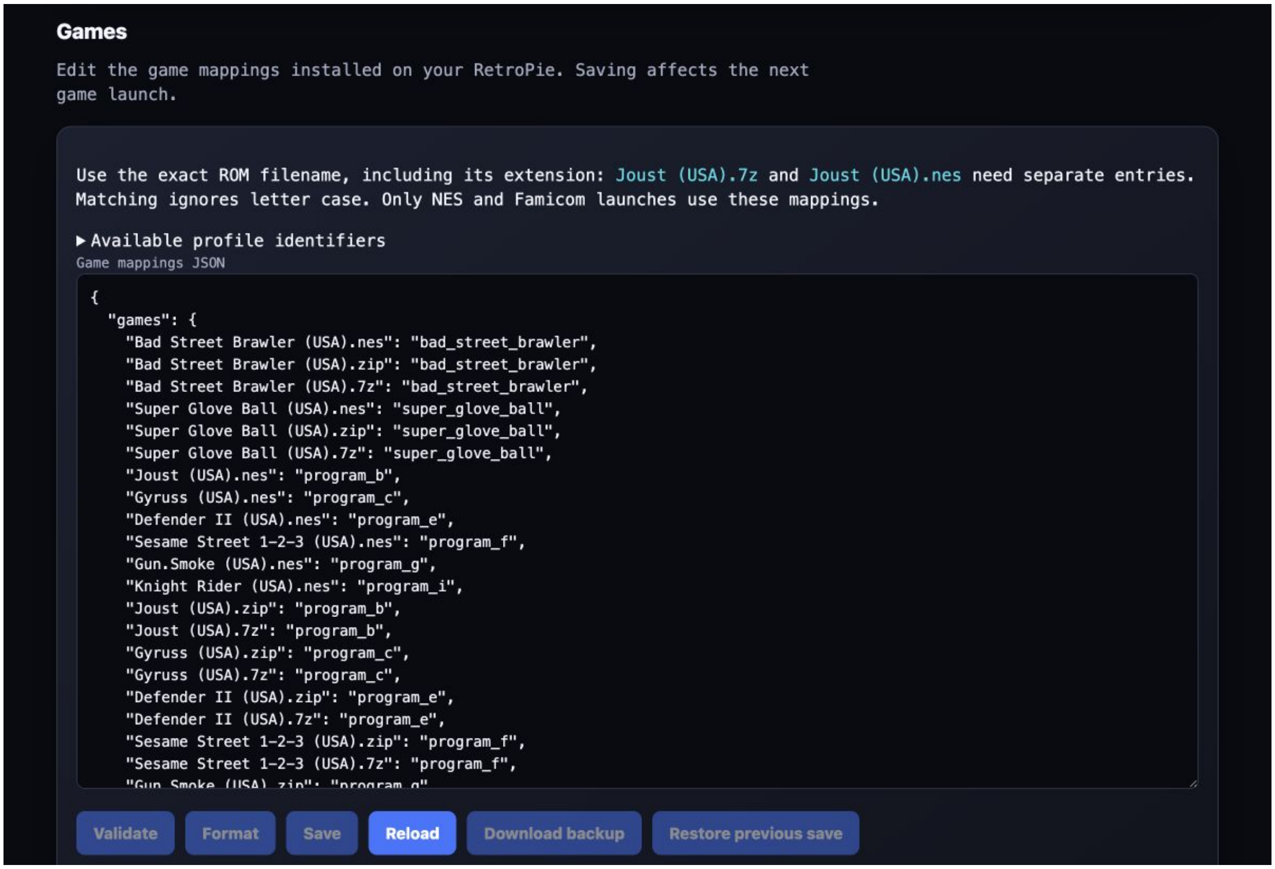
The screenshots below show the current interface. Camera imagery is blurred or excluded for privacy.



Glove Academy with Pixel Pal guiding the personalization choices

In **Glove Academy**, switch on **Tune gestures** to adjust sensitivity. The UNO Q shows a scanning **T** during tuning and a matching scanning **L** during ordinary practice. Both modes pause game input.

Pixel Pal first asks what feels wrong, then presents one instruction at a time. Recording starts only after the whole hand has been tracked clearly and steadily; the user presses **I'm ready** and sees a countdown. The wizard previews a conservative adjustment, requires two successful uses and releases plus three neutral seconds, and enables Save only after that check passes. Saved recognition settings apply across profiles. Numerical thresholds, selective reset, manual preview, and the private diagnostic capture live under **Advanced**. Diagnostic video remains on the UNO Q, is deleted after analysis or cancellation, and its downloadable aggregate report contains no pictures or per-frame hand data.



Games editor in the lower part of Setup

Scroll down **Setup** to **Games** to map exact ROM filenames to profiles. Saving affects the next game launch, not the game already running.

Maintain or extend the project

Use the Installation Guide's maintenance section for updates, and the Configuration Reference for settings and all command options. The Contributing guide covers tests, documentation, package verification, and releases. Printable editions are stored in output/pdf/.

PowerGlove Vision is an independent project licensed under the MIT License. The modified Nestopia core is GPLv2 software and is documented separately from the MIT application; see its distribution and license record. Nintendo, NES, Power Glove, and the named games belong to their respective owners. Third-party software and models retain their own terms, documented in Third-party components.